A nonzero representation V of A is said to be **irreducible** if its only subrepresentations are 0 and V itself, and it is said to be **indecomposable** if it cannot be written as a direct sum of two nonzero subrepresentations. Obviously, irreducible implies indecomposable, but not vice versa.

Typical problems of representation theory are as follows:

(1) Classify irreducible representations of a given algebra A.

(2) Classify indecomposable representations of A.

(3) Do (1) and (2) restricting to finite dimensional representations.

As mentioned above, the algebra A is often given to us by generators and relations. For example, the universal enveloping algebra U of the Lie algebra $\mathfrak{sl}(2)$ is generated by h, e, f with defining relations

$$he - eh = 2e, \quad hf - fh = -2f, \quad ef - fe = h. \tag{2.1.1}$$

This means that the problem of finding, say, N-dimensional representations of A reduces to solving a bunch of nonlinear algebraic equations with respect to a bunch of unknown $N \times N$ matrices, for example system (2.1.1) with respect to unknown matrices h, e, f.

It is really striking that such, at first glance hopelessly complicated, systems of equations can in fact be solved completely by methods of representation theory! For example, we will prove the following theorem.

Theorem 2.1.1. *Let $k = \mathbb{C}$ be the field of complex numbers. Then:*

(i) The algebra U has exactly one irreducible representation V_d of each dimension, up to equivalence; this representation is realized in the space of homogeneous polynomials of two variables x, y of degree $d-1$ and is defined by the formulas

$$\rho(h) = x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}, \quad \rho(e) = x\frac{\partial}{\partial y}, \quad \rho(f) = y\frac{\partial}{\partial x}.$$

(ii) Any indecomposable finite dimensional representation of U is irreducible. That is, any finite dimensional representation of U is a direct sum of irreducible representations.

As another example consider the representation theory of quivers.